Fig.

Function (Mapping or Transformation) :

The mathematical concept of a funtion expresses the intuitive idea that one quantity (the argument of the function, also known as the input) completely determines another quantity (the value or the output). A function assigns exactly one value to each input of a specified type. Since functions are so widely used, many traditions have grown up around their use.The symbol for the input to a funtion is often called the letter x. The symbol for the output is called the dependent variable or value and is represented by the letter y. The function is usually represented by the letter f. Thus the notation $y = f(x)$ indicates that a function named f has an input named x and

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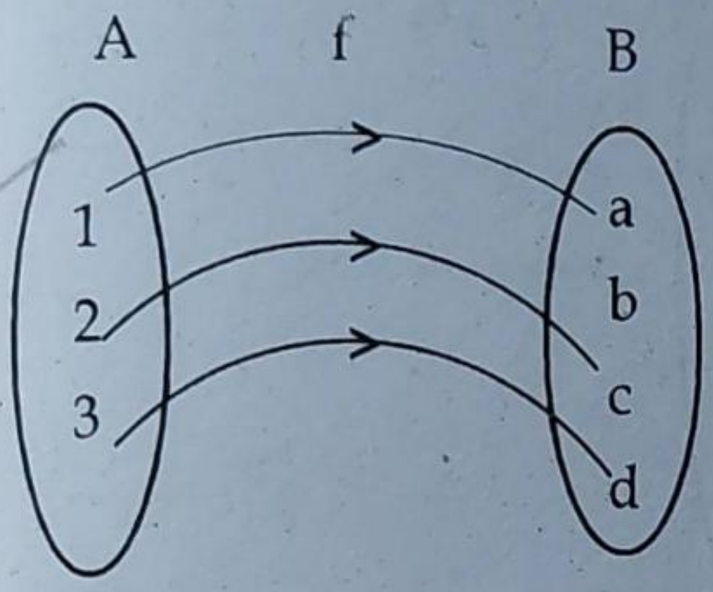
an output named y. The set of all input to a given function is called the domain of the function and the set of all resulting output is called the image or range of the function. The image is often a subset of some larger set, called the condomain of a function.

Let A and B be any two non-empty sets. A relation f from the set A to the set B, written as $f : A \to B$ is called a function if every element of the set A is related to exactly one element of the set B.

Here the set A is called domain and the set B is called codomain.

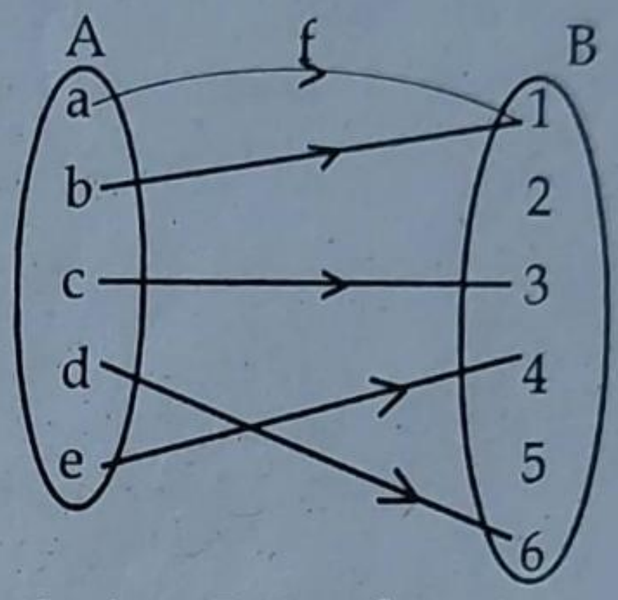
In other words, a function $f : A \to B$ is a set of ordered pairs with the property that for all $x \in A$, there exists a unique $y \in B$ such that $(x,y) \in f$.

(i)



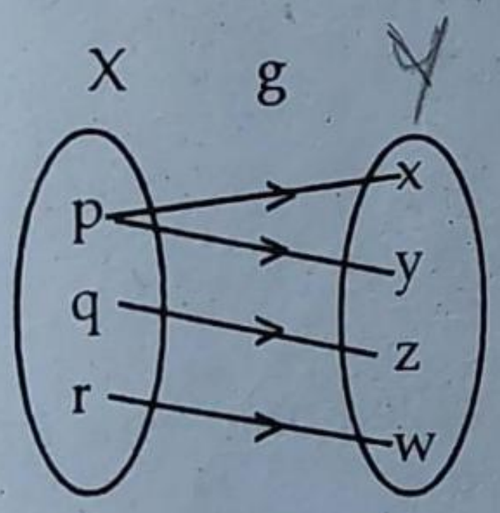
$f : A \to B$ is a function

(ii)



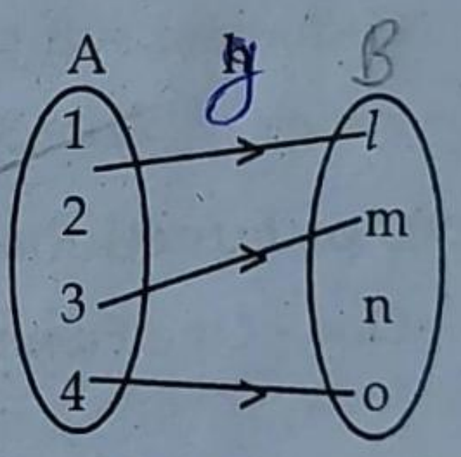
$f : A \to B$ is a function

(iii)



$g : X \to Y$ is not a function as p has two images

(iv)



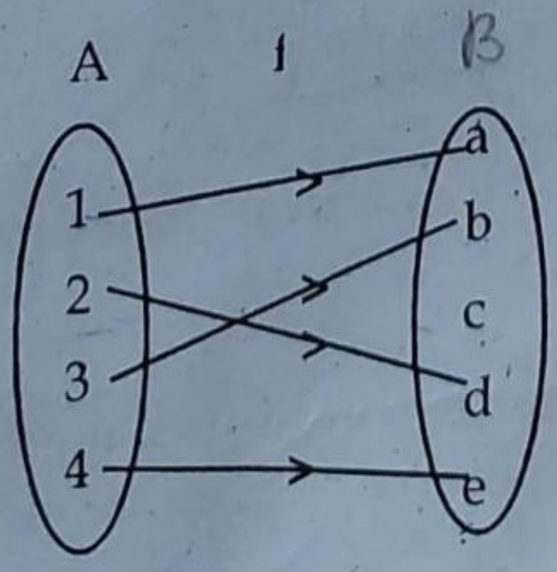
$g : A \to B$ is not a function as 2 does not have an image

Fig

Range : The set consisting of all the images of the elements of A under the function f is called the range of f and is denoted by f (A).

For example

(i)



Fig

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$f : A \to B$ is a function

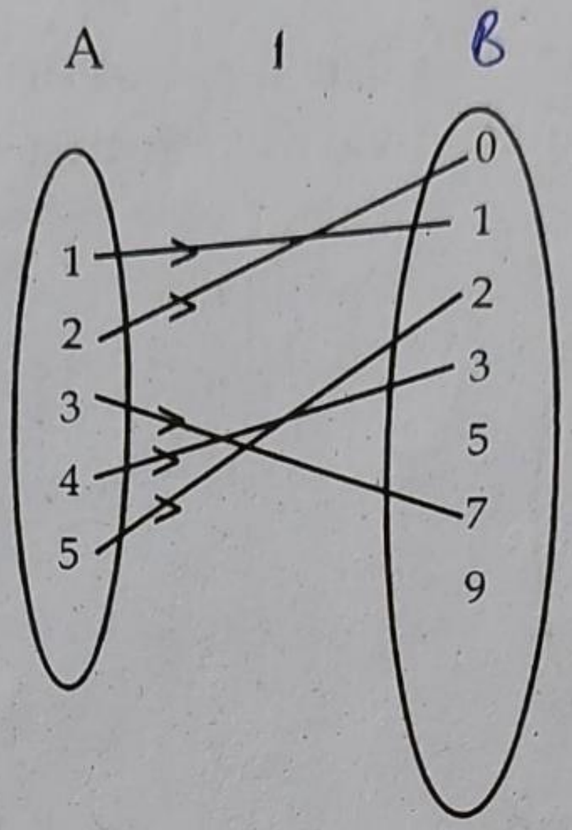
Domain = A = {1,2,3,4}

Codomain = B = {a,b,c,d,e}

Range = {a,b,d,e}

(ii) Let A = {1,2,3,4,5}, B = {0,1,2,3,5,7,9,12,13}, $f : A \to B$,

f = {(1,1), (3,7), (4,3), (5,2)}



Fig

Domain = A { 1,2,3,4,5 }

Codomain = B { 0,1,2,3,5,7,9,12,13 }

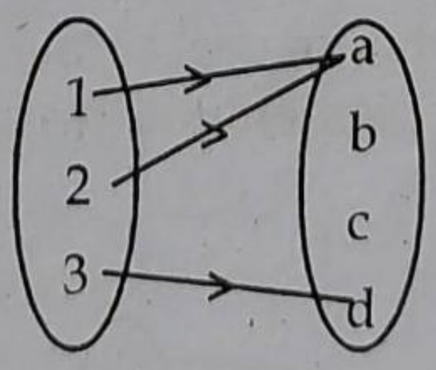
Range = { 0,1,2,3,7 }

2.21 Types of Functions :

(i) **One - One Function :** A function $f : A \to B$ is called one - one (injective or 1-1) if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ or $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$

(ii) **Many - One Function :** A function $f : A \to B$ is called many - one if two or more elements of A have the same image in B.

For example



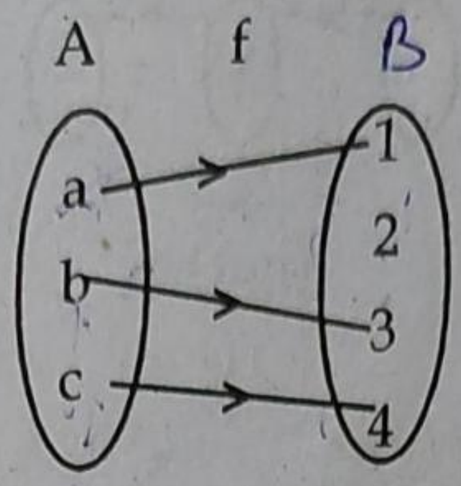
$f : A \to B$ is many one function

Fig

(iii) **Into Function :** A function $f : A \to B$ is called into if there exists atleast one element in B which is not an image of some element in A.

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For example

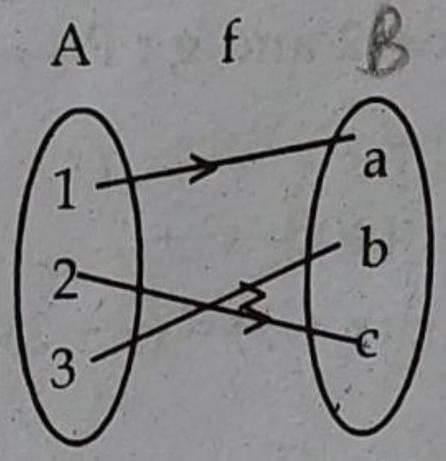


$f : A \to B$ is into function

Fig

(iv) **Onto Function :** A function $f : A \to B$ is called onto (surjective) if every element of B is the image of some element of A. In other words $f: A \to B$ is onto if for every $x \in A$, there exists some $y \in B$ such that $y = f(x)$.

For example



$f : A \to B$ is onto function

Fig

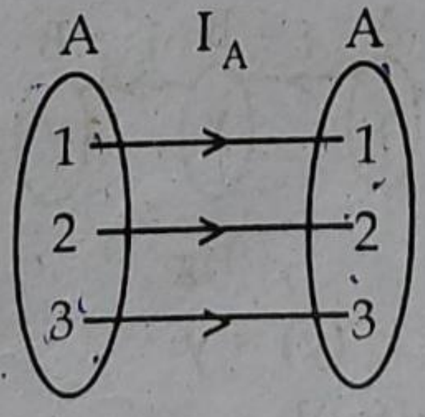
* If $f : A \to B$ is onto function, then

Range of f = Codomain

(v) **Bijective Function :** A function $f : A \to B$ is called bijective if it is both one -one and onto.

(vi) **Identity Function :** A function $f : A \to A$ defined by $f(x) = x$, $\forall x \in A$ is called indentity function and denoted by I_A

For example



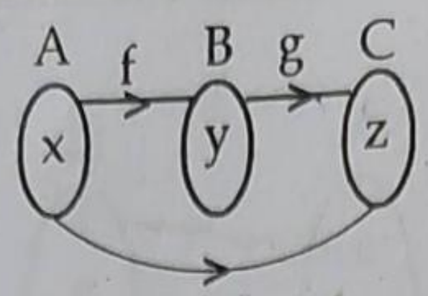
is an identity function

Fig.

2.22 Composition of Functions : Let f be a function from A to B ($f: A \to B$) and g be another function from B to C ($g : B \to C$). Then the composition of f and g denoted by gof (read as g circle f) is a new function from A to C and is given by $(gof)\ x = g\ [f(x)]$

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For example



(gof) x

Fig

Here y = f (x) and z = g (y)

(gof) x = g [f (x)] = g (y) = z

Note :

(i) The Composition of functions is not commutative ie fog $\neq$ gof where f and g are functions.

(ii) gof is called the left composition of g with f.

Example (35) : Let f : R $\rightarrow$ R, f (x) = x + 2 and g : R $\rightarrow$ R, g (x) = x^2 Find fog, gof.

Solution : Given f (x) = x + 2 , g (x) = x^2

(fog) x = f [g(x)] = f (x^2) = x^2 + 2

(gof) x = g [f(x)] = g (x + 2) = $(x + 2)^2$ = x^2 + 4x + 4

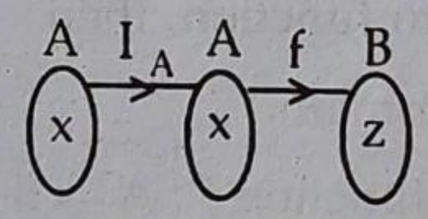
Clearly fog $\neq$ gof

Theorem 1 : The composition of any function with identity function is the function itself

OR

$f \circ I_A = f$ or $I_B \circ f = f$

Proof : **Consider**



Fig

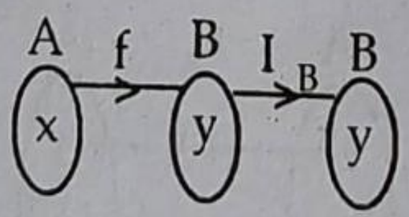
Here I_A (x) = x and y = f (x)

Now (f o I_A) x = f [I_A (x)]

= f (x)

$\therefore$ f o I_A = f

Consider



Fig

Here y = f (x) and I_B (y) = y

Now ($I_B$of) x = I_B [f(x)]

= I_B (y)

= y

= f(x)

$\therefore$ $I_B$of = f

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2.23 Inverse Function :

Let $f : A \to B$ be one - one and onto. Then the inverse function of f is the function that assigns to an element $b \in B$, the unique element $a \in A$ such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence $f^{-1}(b) = a$ whenewer $f(a) = b$.

A function $f : A \to B$ is said to be invertible if the inverse relation $f^{-1} : B \to A$ is also a function.

Note :

(i) In general if $f : A \to B$, then $f^{-1} : B \to A$ need not be a function, it may be a relation.

(ii) If $f : A \to B$ is a bijection (one - one and onto) and $f(a) = b$, then $a = f^{-1}(b)$ where $a \in A$, $b \in B$.

Theorem 2 : If $f : A \to B$ and $g : B \to C$ are one-one and onto functions, then $(gof)^{-1} = f^{-1}og^{-1}$ **(S-06,08,09,W-06)**

Solution : **Given :** $f : A \to B$ and $g : B \to C$ are one-one and onto functions

To prove : $(gof)^{-1} = f^{-1}\, og^{-1}$

Since $f : A \to B$ and $g : B \to C$ are one-one and onto .

$\therefore$ $gof : A \to C$ is also one-one and onto

$\Rightarrow (gof)^{-1} : C \to A$ is also one-one.

Also $g^{-1} : C \to B$ and $f^{-1} : B \to A$ are one-one and onto

$\Rightarrow f^{-1}og^{-1} : C \to A$ is one-one and onto

Let $c \in C$, then there exists an element $b \in B$ such that $g(b) = C$

$\Rightarrow b = g^{-1}(c)$

Also there exists an element $a \in A$ such that $f(a) = b$

$\Rightarrow a = f^{-1}(b)$

Now $(gof)\, a = g[f(a)] = g(b) = c$

$\Rightarrow a = (gof)^{-1}\, c$ - - - - - - - (1)

Also $(f^{-1}og^{-1})\, c = f^{-1}[g^{-1}(c)] = f^{-1}(b) = a$ - - - - - - - (2)

From (1) and (2), we get

$(gof)^{-1}\, c = (f^{-1}og^{-1})\, c$

$\therefore$ $(gof)^{-1} = f^{-1}og^{-1}$

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Example (36) : Let $f : R \to R$, $f(x) = x^3 + 1$, $\forall x \in R$. Then prove that f is a bijective function.

Solution : **Given :** $f : R \to R$, $f(x) = x^3 + 1$, $\forall x \in R$

To prove : f is a bijective function

i.e. f is one-one and onto

One - One : Let $x_1, x_2 \in R$

if $f(x_1) = f(x_2)$

$\Rightarrow x_1^3 + 1 = x_2^3 + 1$

$\Rightarrow x_1^3 = x_2^3$

$\Rightarrow x_1 = x_2$

$\therefore$ f is one-one

Onto : Let $y \in R$ (condomain), then there exists $x \in R$ such that

$f(x) = y$

i.e. $x^3 + 1 = y$

$\therefore$ f is onto

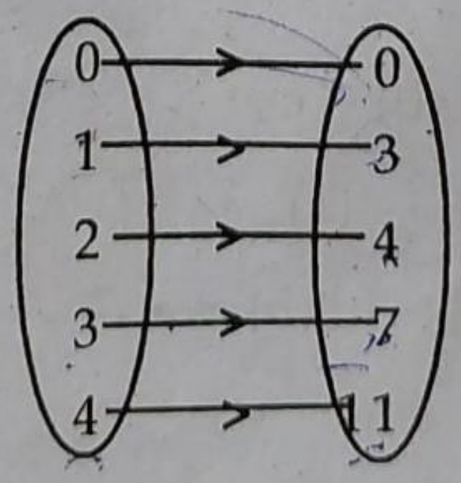
$\therefore$ f is a bijective function.

Example (37) : A function is defined on the set of intengers as follows.

$$f(x) = \begin{cases} x & ; \ 0 \le x < 1 \\ x+2 & ; \ 1 \le x < 3 \\ 4x-5 & ; \ 3 \le x < 5 \end{cases}$$

Find the domain and range and check whether it is one-one or many-one.

Solution : **Given :** $f(x) = \begin{cases} x & , \ x = 0 \\ x+2 & , \ x = 1,2 \\ 4x-5 & , \ x = 3,4 \end{cases}$



Fig

Domain = {0,1,2,3,4}

Range = {0,3,4,7,11}

This function is one-one.

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2.24 **Characteristic Function :** Let U be the universal set and A be any subset of U. Then the Characteristic function f_A of A is defined by

$$f_A(x) = \begin{cases} 1 & , \text{if } x \in A \\ 0 & , \text{if } x \notin A \end{cases}$$

Properties of Characteristic Function :

(i) $f_{A\cap B} = f_A \cdot f_B$ (ii) $f_{A\cup B} = f_A + f_B - f_{A\cap B}$

(iii) $f_{A\cap(B\cap C)} = f_{(A\cap B)\cap(A\cap C)}$ (iv) $f_{A'} = 1 - f_A$

Example (38) : Using the properties of characteristic function, show that

(i) $A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$ (W-08,S-11)

(ii) $A + (B + C) = (A + B) + C$ (W-06,S-08)

(iii) $f_{(A\cup B)}(x) = f_A(x) + f_B(x) - f_A(x).\, f_B(x)$ (S-06)

(iv) $(A')' = A$ (W-08)

Soluiton : (i) To Prove : $A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$

We know that $f_{A\cap B} = f_A \cdot f_B$

$$\therefore f_{A\cap(B\cup C)} = f_A \cdot f_{B\cup C}$$

$$= f_A\,[f_B + f_C - f_{B\cap C}]$$

$$= f_A\, f_B + f_A\, f_C - f_A \cdot f_{B\cap C}$$

$$= f_{A\cap B} + f_{A\cap C} - f_{A\cap(B\cap C)}$$

$$= f_{A\cap B} + f_{A\cap C} - f_{(A\cap B)\cap(A\cap C)}$$

$$f_{A\cap(B\cup C)} = f_{(A\cap B)\cup(A\cap C)}$$

$$\therefore A\cap(B\cup C) = (A\cap B)\cup(A\cap C)$$

(ii) To prove : $A + (B+C) = (A+B) + C$

i.e. $A\cup(B\cup C) = (A\cup B)\cup C$

$$\text{LHS} = f_{A\cup(B\cup C)}$$

$$= f_A + f_{B\cup C} - f_{A\cap(B\cup C)}$$

$$= f_A + f_B + f_C - f_{B\cap C} - f_A \cdot f_{B\cup C}$$

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$$= f_A + f_B + f_C - f_B \cdot f_C - f_A \cdot [f_B + f_C - f_{B \cap C}]$$
$$= f_A + f_B + f_C - f_B \cdot f_C - f_A \cdot [f_B + f_C - f_B \cdot f_C]$$
$$= f_A + f_B + f_C - f_B \cdot f_C - f_A \cdot f_B - f_A \cdot f_C + f_A \cdot f_B \cdot f_C \text{ -------(1)}$$

RHS $= f_{(A \cup B) \cup C}$

$$= f_{A \cup B} + f_C - f_{(A \cup B) \cap C}$$
$$= f_A + f_B - f_{A \cup B} + f_C - f_{A \cup B} \cdot f_C$$
$$= f_A + f_B - f_A \cdot f_B + f_C - f_C \cdot [f_A + f_B - f_{A \cap B}]$$
$$= f_A + f_B - f_A \cdot f_B + f_C - f_C \cdot [f_A + f_B - f_A \cdot f_B]$$
$$= f_A + f_B + f_C - f_A \cdot f_B - f_A \cdot f_C - f_B \cdot f_C + f_A \cdot f_B \cdot f_C \text{ ------ (2)}$$

From (1) and (2), we get $f_{A \cup (B \cup C)} = f_{(A \cup B) \cup C}$

$\therefore A \cup (B \cup C) = (A \cup B) \cup C$

$\therefore A + (B + C) = (A + B) + C$

(iii) To prove : $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_{(A \cap B)}(x)$

Let $x \in A \cup B$, then $f_{A \cup B}(x) = 1$ ------ (1) (by definition)

But if $x \in A \cup B$, then

$x \in A$ or $x \in B$

i.e. $f_A(x) = 1$, $f_B(x) = 1$

If $x \in A \cap B$, then $f_{A \cap B}(x) = 1$

Now $f_A(x) + f_B(x) - f_{A \cap B}(x) = 1 + 1 - 1 = 1$ ------ (2)

From (1) and (2), we get $f_{A \cap B}(x) = f_A(x) + f_B(x) - f_{A \cap B}(x)$

(iv) To prove : $(A')' = A$

$$f_{(A')'} = 1 - f_{A'}$$
$$= 1 - [1 - f_A]$$
$$= 1 - 1 + f_A$$
$$= f_A$$

$\therefore f_{(A')'} = f_A$

$\therefore (A')' = A.$

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Example (39) : List all possible functions from the set X = {a,b,c} to the set Y = {0,1}. Indicate in each case whether the function is one-one or onto or both.(S-07,08,09)

Solution : **Given :** X = {a,b,c} and Y = {0,1}

Let $f : X \to Y$

Here there are $2^3 = 8$ possible distinct functions from X to Y.

(i) $f_0 = f_{000} = \{(a,o), (b,o), (c,o)\}$

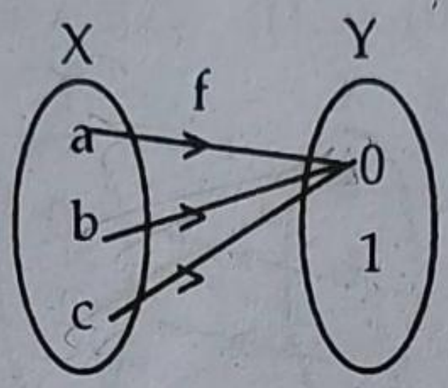
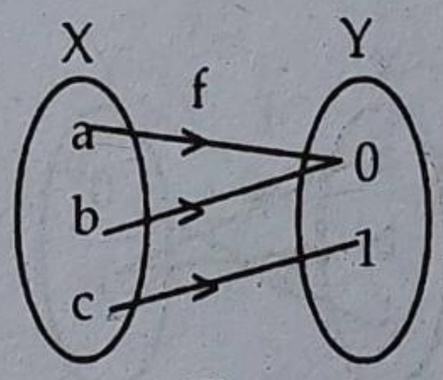


Fig.

From the figure it is clear that $f : X \to Y$ is many-one function and into funtcion.

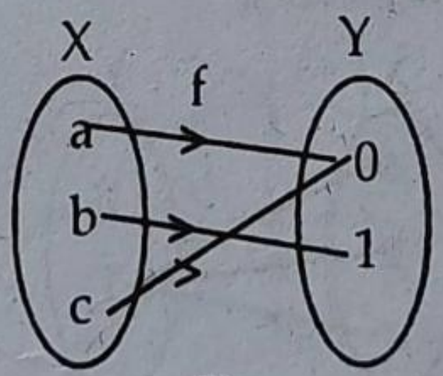
(ii) $f_1 = f_{001} = \{(a,0), (b,0), (c,1)\}$



Fig

From the figure it is clear that $f : X \to Y$ is many-one function and onto function.

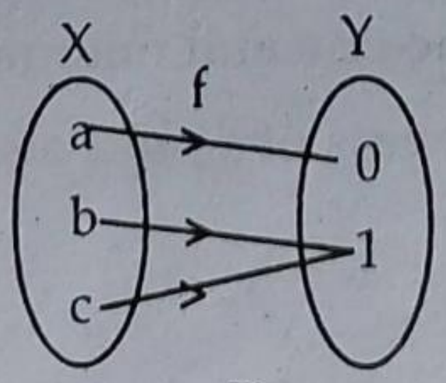
(iii) $f_2 = f_{010} = \{(a,0), (b,1), (c,0)\}$



Fig

From the figure it is clear that $f : X \to Y$ is many-one function and onto function.

(iv) $f_3 = f_{011} = \{(a,0), (b,1), (c,1)\}$

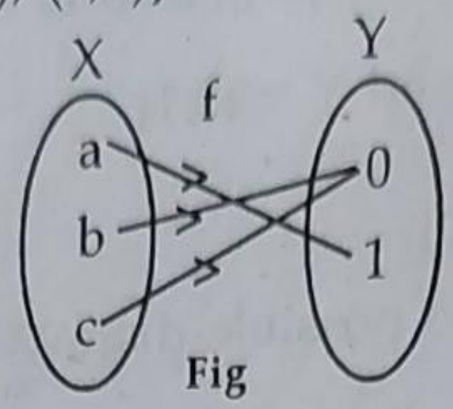


Fig

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From the figure it is clear that $f : X \to Y$ is many-one function and onto function.

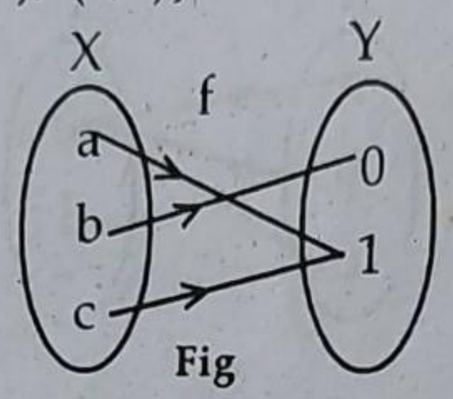
(v) $f_4 = f_{100} = \{(a,1), (b,0), (c,0)\}$



Fig

From the figure it is clear that $f : X \to Y$ is many-one function and onto function.

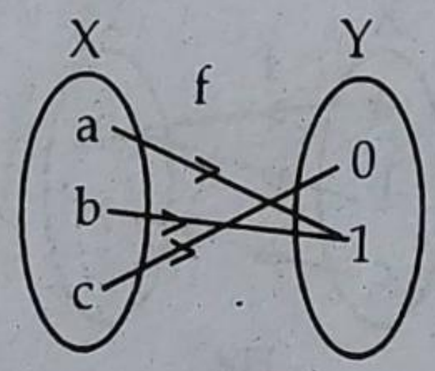
(vi) $f_5 = f_{101} = \{(a,1), (b,0), (c,1)\}$



Fig

From the figure it is clear that $f : X \to Y$ is many-one function and onto function.

(vii) $f_6 = f_{110} = \{(a,1), (b,1), (c,0)\}$



Fig

From the figure it is clear that $f : X \to Y$ is many-one function and into function.

Example (40) : Let f be the set of all one-one and onto mappings from X to X, where $X = \{1,2,3\}$. Find all elements of f and also find inverse of each element. (W-07,S-09)

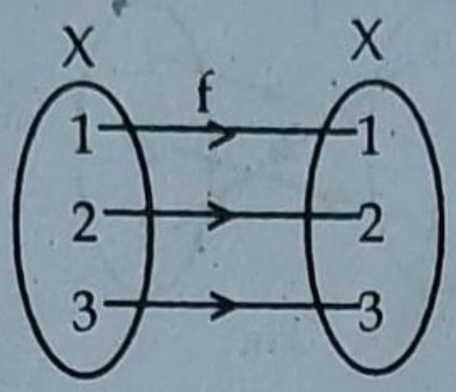
Solution : Given $X = \{1,2,3\}$ and $f : X \to X$

Sine X has three elements

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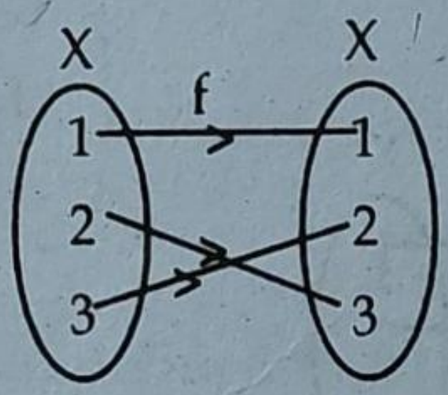
∴ Number of one-one and onto mappings from X to X = 3! = 3×2×1 = 6

(i) f_1={(1,1), (2,2), (3,3)}



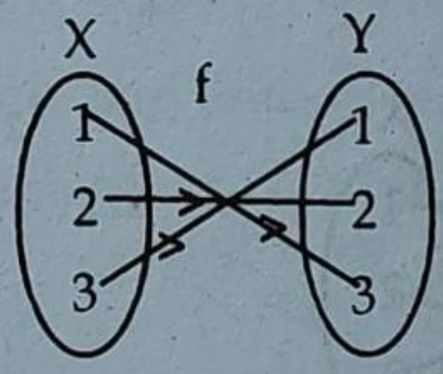
Fig

(ii) f_2 = {(1,1), (2,3), (3,2)}



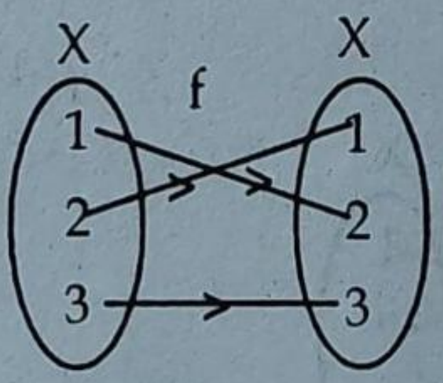
Fig

(iii) f_3 = {(1,3), (2,2), (3,1)}



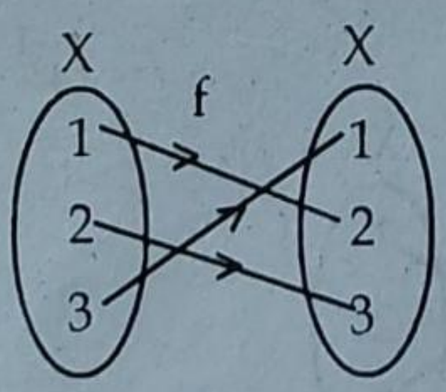
Fig

(iv) f_4 = {(1,2), (2,1),(3,3)}



Fig

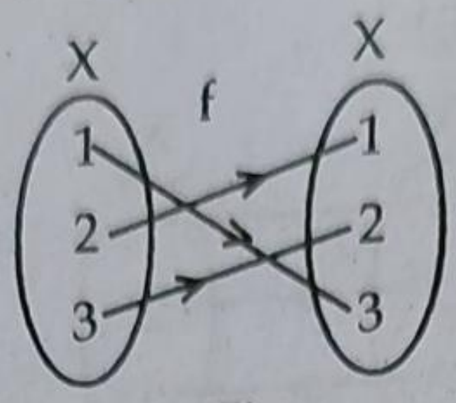
(v) f_5 = {(1,2), (2,3), 3,1)}



Fig

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(vi) $f_6 = \{(1,3), (2,1), (3,2)\}$



Fig

$\therefore$ $f = \{f_1, f_2, f_3, f_4, f_5, f_6\}$

Now we will find the inverse of each element.

(i) f_1^{-1} $\therefore$ $f_1^{-1} = f_1 = I$

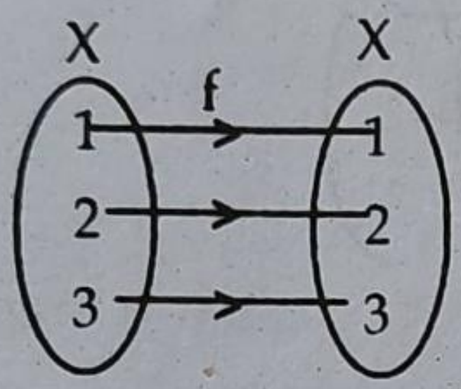


Fig.

(ii) f_2^{-1} $\therefore$ $f_2^{-1} = f_2$

Fig.

(iii) f_3^{-1} $\therefore$ $f_3^{-1} = f_3$

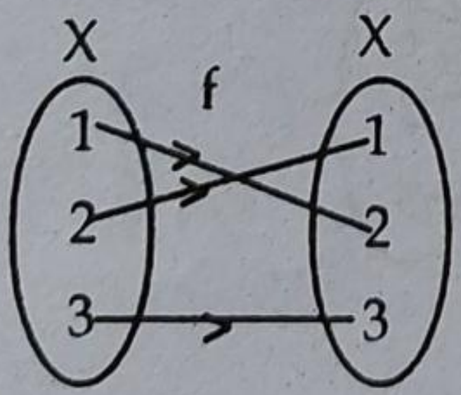


Fig.

(iv) f_4^{-1} $\therefore$ $f_4^{-1} = f_4$

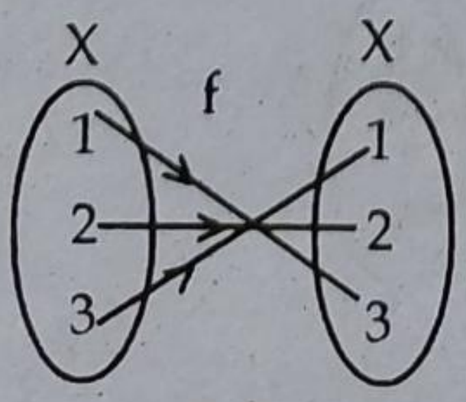


Fig.

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(v) f_5^{-1} $\therefore f_5^{-1} = f_6$

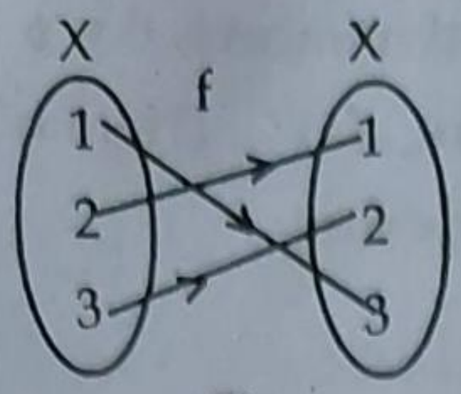


Fig.

(vi) f_6^{-1} $\therefore f_6^{-1} = f_5$

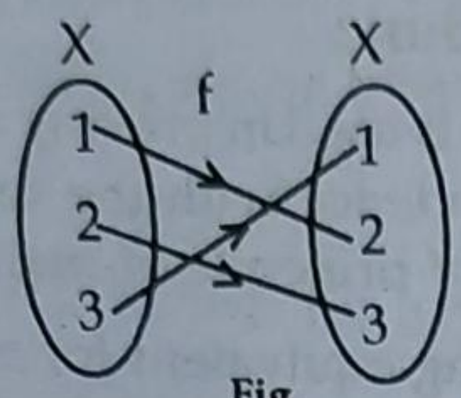


Fig.

Example (41) : Show that the function defined by $f : Q \to Q$ such that $f(x) = 3x+4$ for all $x \in Q$ is one-one and onto where Q is the set of rational numbers. Also find a formula that defines the inverse function f^{-1} (W-09)

Solution : Given : $f : Q \to Q$ such that $f(x) = 3x+4$

To prove : f is one-one and onto

(i) Let $x_1 . x_2 \in R$ such that $f(x_1) = f(x_2)$

i.e. $3x_1 + 4 = 3x_2 + 4$

i.e. $3x_1 = 3x_2$

i.e. $x_1 = x_2$

$\therefore f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

$\therefore$ f is one-one

(ii) Let $y \in Q$ such that $y = f(x)$

i.e. $y = 3x + 4$

i.e. $3x = y - 4$ ie $x = \frac{1}{3}(y - 4)$

Hence for a given $y \in Q$, there exists an element $x = \frac{1}{3}(y - 4) \in Q$

Such that $f(x) = y$

$\therefore$ f is onto

Since f is one-one and onto

$\therefore$ f is invertible

$\therefore f_y^{-1} = \frac{1}{3}(y - 4)$

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